

Probing Pretraining “Quality”: Sociolinguistic Mobility as a Lens on Web Corpus Filters

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Abstract

Data selection algorithms for large language model (LLM) pretraining are designed to mine ‘quality’ from the web, but that notion is murky at best. This study explores whether sociolinguistic mobility – realized through language use, in context – offers a lens for understanding what such algorithms actually select. As a case study, the FineWeb pretraining pipeline is examined for how it reshapes two tangible axes of language use: the geographical provenance and register composition of texts drawn from ICE and GloWbE. FineWeb rejection rates are highest, and fairly uniform across provenance groups, for web-native and high-context registers (90.9% and 99.2%, respectively) but within medium-context registers, association with the United States is moderately protective: the 7% rejection rate is well below the medium-context average of 38%, an effect corroborated by NPMI. Unlike prior work on learned, classifier-based filters and US-internal sociolect bias, this study finds comparable effects in a deterministic pipeline within the global spread of English. As pretraining data selection becomes increasingly opaque, these results suggest that algorithmic ‘quality’ does not represent a real-world range of English use manifested in the web-as-corpus – a gap made visible by tangible sociolinguistic dimensions.

1 Introduction

The construction of LLMs today relies on massive web-scraped archives, popularly perceived as ‘raw’ data from which objectively valuable knowledge can be extracted, despite growing critical scholarship that complicates this framing (Baack, 2024; Desai et al., 2024). This epistemological assumption has motivated the deployment of various data curation protocols to sanitize LLM pretraining input, including a range of ‘quality’ filters (Albalak et al., 2024). Indeed, the world-wide web hosts myriad linguistic material (Crystal, 2011; Blommaert, 2018; Mair, 2025) of varying utility for NLP

tasks – and yet, the manner in which data curation manipulates the “maturing medium” (Tinnefeld, 2023) of internet language has eluded full capture. Sociolinguistic perspectives on language modeling provide insight into this open area of research: “the validity and value of a language model depends on the degree to which the training corpus accurately represents the variety that is effectively being modeled [...] even if that variety of language is unknown or under-specified” (Grieve et al., 2025, p. 5). This raises the question of which dimensions of linguistic diversity matter most as LLM-based technologies are exposed to an increasing number of social and communicative contexts.

To that end, this work begins by presenting a paradigm of sociolinguistic *mobility*, following Blommaert (2010), as a lens to describe the variation in English, manifested on the web-as-corpus (§2). This motivates the central study, wherein the FineWeb quality filtering pipeline (Penedo et al., 2024) is applied to a corpus compiled from ICE and GloWbE, spanning a range of geographic and register contexts (§3). The results show that register context level is strongly associated with filter rejection, with a further interaction between register and geographic provenance at the medium-context level (§4), suggesting that sociolinguistic axes can help describe algorithmic behavior. §5 discusses directions for future work, including more granular analysis of the low-context subset, where rejection patterns are more volatile and provenance-level trends remain preliminary. Experimental code will be made available on GitHub¹.

2 Sociolinguistic Mobility and Web-as-Corpus

The prevailing paradigm for linguistic variation centers on the *distribution* of language varieties in

¹<https://github.com/jessiclassy/pretraining-quality-mobility>

place and time (Kohnen and Mair, 2012; Trudgill, 2017; Kortmann et al., 2020; Szmrecsanyi and Grafmiller, 2023a), characterized by a range of features (e.g., phonological, lexical and/or morphosyntactic). In effect, this paradigm treats varieties as fixed, discrete classes to be identified and balanced across a dataset, as if representation were a classic machine learning problem. This framing has been widely embraced by contemporary dialect-centered NLP approaches (Joshi et al., 2025) and continues to inform directions for language technology development (Bird, 2022), such as language-variety benchmarks built via synthetic data generation, simulating regional dialects as ‘translations’ of standard English (Ziems et al., 2022, 2023; Lee et al., 2025; Oh et al., 2026). But treating varieties as classes to be balanced takes them as a given, when they have evolved (and continue to develop) from ongoing geographic and sociohistorical processes: contact, migration, standardization.

A more recent paradigm, emerging from works such as Pennycook (2006) and Blommaert (2010), reframes language varieties as one among several tangible resources, including register and genre, which are continually produced and reused as speakers move through the social, spatial and temporal trajectories of their lives. Blommaert makes the stakes of this shift explicit:

The crux of the matter is that we need to think of issues such as linguistic inequality as being organized around concrete resources, not around languages in general but specific registers, varieties, genres. And such concrete resources follow the predicament of their users: when the latter are socially mobile, their resources will follow this trajectory; when they are socially marginal, their resources will also be disqualified. [...] If we fail to do that, we risk drawing a caricature of social realities and becoming very upset about that caricature rather than about an accurate replica of social processes. (Blommaert, 2010, p. 47)

The distributional paradigm’s flaw, in this view, is not that it attends to variety, but that it treats variation as a static inventory rather than the product of ongoing sociohistorical processes. This inadvertently ignores how a linguistic resource may be disqualified: users who are socially marginal have their resources disqualified alongside them, a

Provenance group	Regions
1 (Traditional L1)	United States, Great Britain
2 (High-contact L1)	Australia, Canada, Ireland, New Zealand
3 (Indigenized L2)	Singapore, Hong Kong, India, Philippines,
4 (High-contact L2)	Jamaica, Kenya, Nigeria, Tanzania

Table 1: Geographic provenance, grouped by sociohistorical patterns of English adoption.

concern raised over a decade ago and still echoed by sociolinguists writing on LLMs (Grieve et al., 2025; Nguyen, 2025; Lutgen et al., 2026).

Pretraining data filters risk enacting a comparable disqualification: each filtering decision determines which linguistic resources move forward into a corpus, a model, and its downstream outputs – and which do not. Capturing this process requires attention to at least two things distribution leaves flat: the sociohistorical conditions from which a resource emerges, and the situational context in which it is deployed. This study operationalizes the former as geographic provenance and the latter as register context level, examined jointly.

English is a strategic case for this kind of analysis: no other language is as widely used on the web, nor carries as much internal variation as a consequence of its complex history of spread (Kortmann et al., 2020; Pimienta, 2022). This makes it an unusually rich site for observing how sociolinguistic resources are manipulated in the data that trains today’s language models.

3 Probing Pretraining Filters

To probe at the notion of pretraining ‘quality’ in terms of geographic provenance and register context level, this study applies the open-source FineWeb pretraining filter pipeline (Penedo et al., 2024) to texts drawn from existing English corpora. Resulting document rejection rates are analyzed across both dimensions of sociolinguistic variation as a practical measure of web corpus filter behavior, supplemented by normalized pointwise mutual information.

3.1 Representing Sociolinguistic Mobility

3.1.1 Existing corpora

Data from fourteen nations with attested development of local English dialects are drawn from two

Context level	Examples
low	printed academic writing, legal presentations
medium	personal correspondence, scripted conversations, press editorials, instructional/creative writing
high	private/unscripted conversations and monologues

Table 2: Register context levels, grouped by situational characteristics, such as participation, channel, production circumstances, and setting.

existing corpora: the Corpus of Global Web-based English (Davies and Fuchs, 2015), abbreviated as GloWbE, and the International Corpus of English (Greenbaum, 1991; Kirk and Nelson, 2018), abbreviated as ICE. GloWbE was collected across a six-month period of global usage of web-native English, scraping over 1.5M web pages from Google’s search index. ICE, by contrast, consists entirely of non-web documents, but is uniquely detailed in its geographical and situational annotation. Over the past two decades and across more than a dozen academic institutions, teams of linguists have manually curated each regional component to capture approximately one million words of prototypical, local English, targeting 60% transcribed speech and 40% written texts. All regional components follow a shared annotation schema to flag editorial information (e.g., overlapping speech, normative corrections), as well as a document genre typing (e.g., scripted or unscripted speech, academic writing, editorials, correspondence). We posit that this representation makes an important proxy for web text types such as institutional/professional texts, podcast or broadcast transcripts, or even the disfluent, multi-turn language found in web forums, chat logs or comment sections. Both existing corpora are considered essential resources to large-scale and fine-grained research on English variation (Mair, 2017; Szmrecsanyi and Grafmiller, 2023b; Qin, 2023; Collins, 2024; Basile, 2025). Thus, the dataset resulting from combining both corpora enables a web-scale analysis of how filter behavior is affected by geographical provenance, as well as a small-scale analysis of register context level, both independently and in combination with provenance.

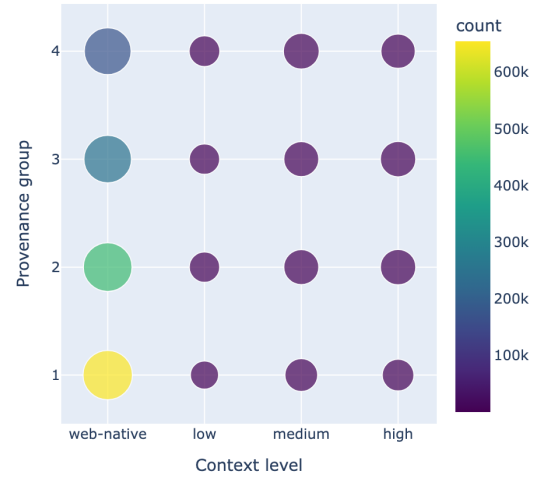


Figure 1: Data sizes with respect to provenance group and register context level.

3.1.2 Preprocessing

To align these documents with FineWeb’s expected plaintext format, a registry of regular expressions is compiled to strip $> 99.9\%$ of ICE-specific intra-document feature annotations so that only the naturalistic content remains. No punctuation marks are stripped or added to ICE transcribed speech. Similarly, residual HTML tags and fair-use masking strings are removed from GloWbE documents (see Appendix C). Finally, both corpora are adapted to JSONL format.

To enable cross-corpus analysis of geographical provenance, all documents are grouped as described in Table 1, derived from the sociohistorical patterns in which English was adopted in a specific region (Trudgill, 2017; Kortmann et al., 2020). ICE-specific register labels are also categorized into three coarse situational-context categories: low, medium, high, as shown in Table 2. These categories are assigned through a qualitative grouping of ICE genre labels by the level of situational context needed between participants, guided by four of the characteristics identified in Biber and Conrad (2009)’s descriptive framework of situational register: participants, channel, production circumstances, and setting. In the absence of curated register labels for GloWbE documents, they are analyzed as a whole under a general ‘web-native’ category whose situational context is under-determined.

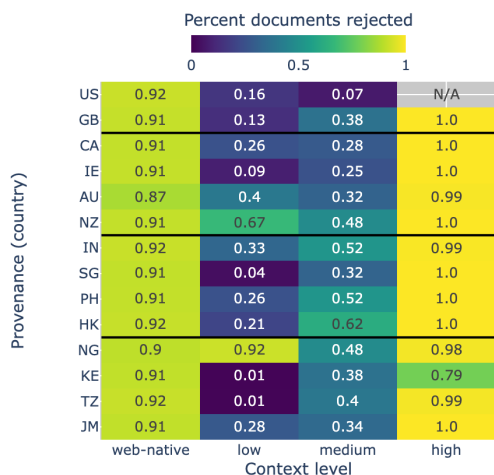


Figure 2: Heatmap of reject rates, aggregated by geographical provenance and register context levels. Solid lines indicate provenance groups.

3.2 Analyzing the FineWeb Quality Pipeline

For this experiment, the combined English corpus is processed through the five stages of FineWeb pipeline (Penedo et al., 2024) to obtain per-document rejection decisions, which are aggregated over the two sociolinguistic axes for analysis. These stages cover English language identification, repetition filters, and several deterministic quality filters which screen documents for specific character/word/line thresholds. Filtering algorithm details can be found in Appendix B.

4 Results & Discussion

Figure 2 presents a heatmap of document rejection rates, aggregated by geographical provenance and register context levels for all fourteen countries represented in both GloWbE and ICE. Broadly speaking, all provenance groups follow the same trend: web-native GloWbE subsets pattern with high-context texts, both showing the highest overall rejection rates, averaging at 90.9% and 99.2% respectively. Within medium-context texts, association with the United States appears to have a moderately protective effect, with a 7.1% rejection rate compared to the average medium-register rate of 38.4%. This is corroborated by normalized pointwise mutual information ($\text{NPMI} \in [-1, 1]$, where 0 indicates independence and negative values indicate avoidance), shown in Figure 3 ($\text{NPMI} = -0.33$). The FineWeb pipeline largely discards resources

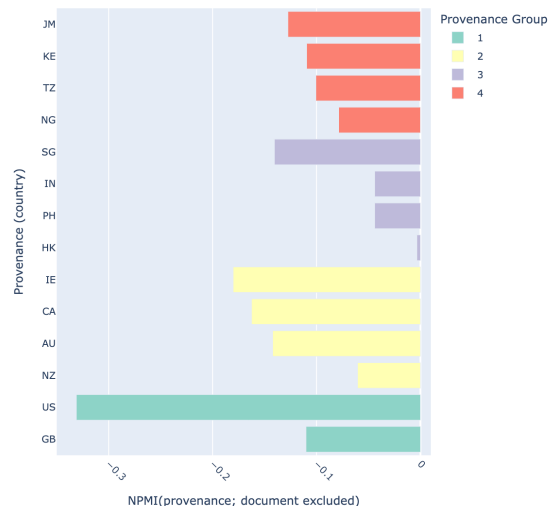


Figure 3: Normalized pointwise mutual information score for geographical provenance and filter rejection, within the medium register context level.

of primary importance to corpus linguistics and sociolinguistic traditions, reproducing the dominance of mainstream, US-centered linguistic contexts. This pattern emerges from an algorithmic mechanism distinct from the classifier-based biases identified in prior work (Dodge et al., 2021; Gururangan et al., 2022; Lucy et al., 2024; Deas et al., 2025). In other words, the notion of ‘quality’ remains divorced from a real-world range of language use.

Rejection rates for low-context texts are much more volatile, and are not analyzed in depth here. More than half of the geographies represented among low-context texts are more protected from filter rejection than their medium-context counterparts by at least 10%, though patterns vary considerably by country: association with the United States, Canada, Australia, and Jamaica yield similar rejection rates to their medium-context counterparts, while association with Nigeria and New Zealand yield far greater ones. Given the small size of this low-context subset ($11 \leq n \leq 69$ for each country), these patterns are best treated as preliminary rather than conclusive, with a more granular analysis of low-context text left to future work.

5 Conclusion

This study finds that provenance and register, examined jointly, reveal structure in FineWeb’s filtering behavior that neither dimension shows on its

own: rejection is strongly associated with register context. Within medium-context text specifically, association with the United States is moderately protective against rejection, an effect that attending to language variety alone, without the sociohistorical conditions of its production or situational use, would not have surfaced this interaction. As the practice of LLM pretraining data curation continues to grapple with what “quality” text means, articulating the dimensions of web-as-corpus through sociolinguistic terminology offers a concrete vocabulary, capable of describing not just what a filter selects, but what kind of language use it renders legible or illegible in the process.

Beyond the filtering pipeline studied here, this approach could be extended to compare pretraining data curation strategies across filters (Lucy et al., 2024; Deas et al., 2025), or applied to data mixing protocols (Chen et al., 2026), where sociolinguistic sensitivity may operate differently than under binary retention decisions. More broadly, it appears that a resource-oriented, mobility-based perspective may provide an initial step toward a fuller diagnostic vocabulary for evaluation what data-centric curation choices encode about language on the web.

Limitations

Simplicity of the mobility framework. The mobility framework operationalized here reduces a rich theoretical construct to a handful of discrete levels, inferred from corpus provenance and genre/register conventions. This is a deliberately coarse instrument, sufficient to establish whether filtering outcomes are sensitive to mobility at all, but not to capture finer-grained distinctions within or across mobility levels. Future work might operationalize mobility more directly, for instance through document-level features rather than corpus-level proxies.

Parity of ICE and GloWbE. The two corpora unified for this study were built using very different methods, capturing English variation across differing communicative contexts, and resulting in starkly different data volumes. While the present study treats ICE and GloWbE jointly to explore the diversity represented among these texts, any direct comparison of results between the two corpora should be treated with caution, partly because GloWbE lacks register/genre annotation comparable to ICE’s. Future work may leverage automatic register classification labels (Laippala et al.,

2023) to expand upon the ICE-centric results in the present study.

Statistical rigor of rejection patterns. The present study focuses concrete measurements of rejection rates across the dimensions of geographic provenance and register context levels, supported by normalized pointwise mutual information scores to describe the site of interaction between these variables. This approach is especially limited for sparse subsets, such as low-context text, where country-level sample sizes are small (as few as 11 documents) and rejection rates are correspondingly volatile. The interpretation of such results is limited, but points to promising avenues which could be more rigorously examined with mixed effect modeling, as is common in the sociolinguistic tradition (Gudmestad and Metzger, 2025, *inter alia*), or perplexity analyses (Deas et al., 2025; Dunn, 2024).

Treating FineWeb as a black box. This work is oriented towards a single pretraining filtering pipeline as a practical approximation of how filtering preferences are manifested, but in doing so, the pipeline itself is treated as a black box. Such filtering pipelines actually consist of a sequence of deterministic text filters. Detailed characterization of the rejection patterns specific to each stage or specific text heuristic are out of scope for the present study, but the progression of rejection patterns across pipeline stages is available in Appendix B, and an initial breakdown of text heuristic patterns are available in Appendix D.

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A Source Corpora Metadata

Specific document counts for GloWbE and ICE are presented in Tables 3 and 4.

B Filtering Pipeline

The FineWeb filtering pipeline (Penedo et al., 2024) consists of five stages, described below. Data volumes after every stage are presented in Figure 4, and shows that the most aggressive filter is in fact the FineWeb quality filter.

Country	Abbreviation	Web pages
United States	US	275,156
Canada	CA	135,692
Great Britain	GB	381,841
Ireland	IE	102,147
Australia	AU	129,244
New Zealand	NZ	82,679
India	IN	113,765
Singapore	SG	45,459
Philippines	PH	46,342
Hong Kong	HK	43,936
Nigeria	NG	37,285
Kenya	KE	45,962
Tanzania	TZ	41,356
Jamaica	JM	46,748
TOTAL		1,527,612

Table 3: GloWbE metadata

Country	Abbreviation	Written	Spoken	Total
United States	US	200	0	200
Great Britain	GB	200	300	500
Canada	CA	200	300	500
Ireland	IE	200	300	500
Australia	AU	199	300	499
New Zealand	NZ	200	300	500
India	IN	200	300	500
Singapore	SG	200	300	500
Philippines	PH	200	300	500
Hong Kong	HK	200	300	500
Nigeria	NG	504	392	896
Kenya	KE	181	167	348
Tanzania	TZ	175	115	290
Jamaica	JM	200	299	499
TOTAL		3,059	3673	6,732

Table 4: ICE metadata

B.1 Language Identification

This algorithmic filter will exclude documents that do not produce a language identification score $p(\text{eng}) > 0.65$. The language identification model is the `fasttext` classifier (Joulin et al., 2016).

B.2 Gopher Repetition Filter

This algorithmic filter computes the fraction of duplicate lines, paragraphs, and character-level n -grams, and excludes documents when any of those fractions exceed a hardcoded threshold. It was originally deployed to pretrain the Gopher model (Rae et al., 2021).

B.3 Gopher Quality Filter

This algorithmic filter also originates from (Rae et al., 2021), but is defined instead by applying a variety of text heuristics: number of words, mean word length, symbol-to-word ratio, bullet point presence, and stop word presence.



Figure 4: Data volumes after every stage; Stage 0 represents the initial volume before any filtering.

B.4 C4 Quality Filter

This algorithmic filter was deployed to create the prominent pretraining dataset C4 (Raffel et al., 2020), and excludes documents by applying another set of text heuristics: presence of the words “Javascript”, “lorem ipsum”, “terms of service”, or curly brackets. Additionally, all lines without terminal punctuation are penalized such that if a document is found to have fewer than 5 non-penalized sentences or contain non-penalized lines with 3 or fewer words, it is discarded.

B.5 FineWeb Quality Filter

This algorithmic filter was designed specifically for the FineWeb dataset (Penedo et al., 2024), designed to be applied after the preceding filters and maximize aggregate accuracy on several prominent NLU benchmarks, such as HellaSwag (Zellers et al., 2019). In actuality, its definition is based entirely off of deterministic line-punctuation, newline, and character duplicate ratios.

C Source Corpus Examples

For brevity, the source corpus examples below are truncated to the first 1000 characters.

C.1 GloWbE Sample

C.1.1 Cleaned Text

About this item

As stated in the thesis project , " My community consists of over 650 non-profit organizations .These agencies range in size from the small local grassroots organization to fully staffed , fully funded government/private agencies .They consist of Soup Kitchens , Daycares and Senior Centers , M/R and M/H facilities , Drug/alcohol rehabs and Churches of all different denominations .The common thread that these agencies hold is the fact that they are all " social service agencies " that contain a feeding program that serves low-income individuals in some form whether through hot-prepared on-site meals or emergency food packages .The other common thread that they all share is membership into the Greater Philadelphia Food Bank which is a food collection , storage and distribution center . The collective mission of my community is to end " Hunger " .The need as identified through my observation as Agency Relations Director is to strengthen and focus these social service ag...

C.1.2 Original Text

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agencies hold is the fact that they are all " social service agencies " that contain a feeding program that serves low-income individuals in some form whether through hot-prepared on-site meals or emergency food packages . The other common thread that they all share is membership into the Greater Philadelphia Food Bank which is a food collection , storage and distribution center . The collective mission of my community is to end " Hunger " . Each individual organization has its own defined method of achieving that result in conjunction with other goals ..

C.2 ICE Sample

C.2.1 Cleaned Text

Chandler and Frasier: Kampala's Most Wanted | Ernest Bazany

"Well done."

"What?"

The old woman continued to smile broadly, all teeth and good nature. Her eyes were so squinted up with pleasant spirit that only bits of the black parts were visible. "Well done," she repeated, confident that the young man could understand her.

The young man could not. Chandler was a thoroughly and totally and completely worthless schoolboy so he was not used at all to hearing those words, together, in that order. He met the repetition of her statement with the repetition of his own. "What?"

This is when his elder brother Frasier joined in the conversation, brusquely nudging in and scowling. "She means 'Mugyebale' , you doofus

."

Well done was, after all, the exact transliteration of the common Luganda greeting Mugyebale and, this was Kampala, the city where most people spoke first and worried only later, if ever, whether their speech had made any sense. It was also a common thing for tee...

C.2.2 Original Text

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<I> <h> <ICE-UG:W2F-016#1:1>
Chandler and Frasier: <w>
Kampala's </w> Most Wanted |
Ernest Bazanye </h>
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<p> <quote> <ICE-UG:W2F-016#2:1> "Well
done." </quote> </p>
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<p> <quote> <ICE-UG:W2F-016#3:1> "What?"
</quote> </p>
```

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<p> <ICE-UG:W2F-016#4:1> The old woman
continued to smile broadly, all
teeth and good nature. <ICE-UG:W2F-
016#5:1> Her eyes were so squinted
up with pleasant spirit that only
bits of the black parts were visible.
<quote> <ICE-UG:W2F-016#6:1> "Well
done," </quote> she repeated,
confident that the young man could
understand her. </p>
```

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<p> <ICE-UG:W2F-016#7:1> The young man
could not. <ICE-UG:W2F-016#8:1>
Chandler was a thoroughly and
totally and completely worthless
schoolboy so he was not used at all
to hearing those words, together, in
that order. <ICE-UG:W2F-016#9:1> He
met the repetition of her statement
with the repetition of his own. <
quote> <ICE-UG:W2F-016#10:1> "What?"
</quote> </p>
```

```
<p> <ICE-UG:W2F-016#11:1> This is when
his elder brother Frasier joined in
the...
```

D Filter Reason Breakdown

In Figure 5, The high-context facet plot shows 0% rejection for the US column because there is no high-context US data available. In contrast, the low-context facet plot shows 0% rejection for Kenya and Tanzania because none of the 22 legal presentation documents were rejected by FineWeb.

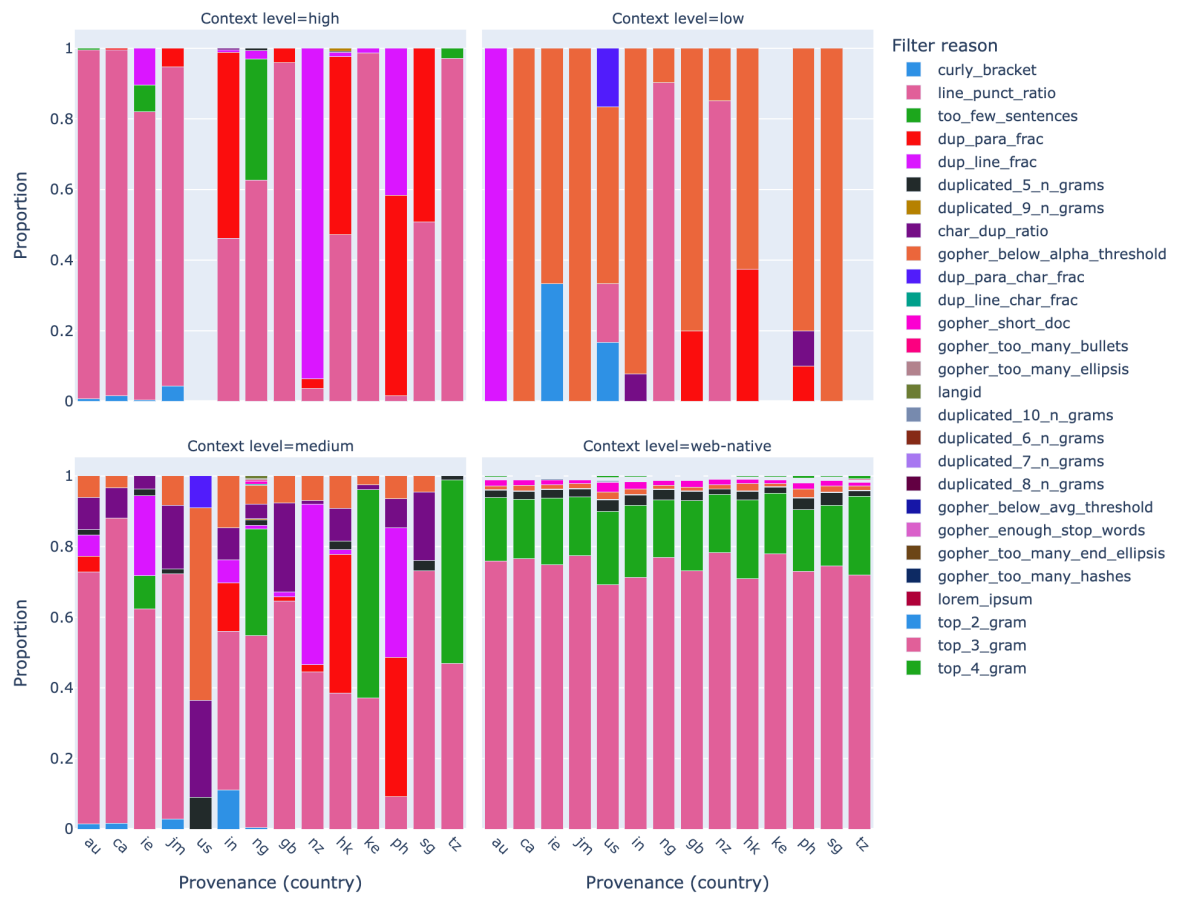


Figure 5: The relative proportions of filter rationales, aggregated by geographic provenance and register context levels.